



Soknarith (Soki) Sem

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SUMMARY

Materials scientist with training in chemical engineering and applied mathematics, specializing in statistical thermodynamics, transport phenomena, and atomistic modeling. Experienced in first-principles quantum mechanics simulations (VASP & GPAW), classical molecular dynamics simulations (LAMMPS), and physics-based modeling to quantify defect energetics, transport behavior, and temperature-dependent properties.

WORK EXPERIENCE

Research Assistant & Graduate Teaching Assistant Worcester Polytechnic Institute, Worcester, MA	May 2024 to Present
- Developed physics-based thermodynamic and transport models for defect-mediated ionic conductivity in Ruddlesden-Popper (RP) phase oxide perovskites - Computed temperature-dependent free energies and defect energetics using first-principle (DFT) simulations - Analyzed structure-property-process relationships governing ionic transport across temperature regimes - Assisted instruction in Applied Analytical Methods (ODEs/PDEs) and Metal Additive Manufacturing	
Chemical Process Engineer Knowles Precision Devices, New Bedford, MA	May 2023 to May 2024
- Led R & D-to-production transition for high-voltage capacitor dielectric fluids - Optimized manufacturing layout and process flow, reducing production costs by 15% - Authored SOPs and technical documentation to support scale-up and cross-team transfer	
R & D Technician Factorial Energy, Woburn, MA	Mar 2023 to May 2023
- Fabricated Li-metal batteries in controlled glovebox and dry-room environments - Reduced material waste by 20% through inventory system redesign	
Purification Engineer Eurofins, Framingham, MA	Jul 2022 to Mar 2023
- Purified therapeutic proteins using AKTA Avant systems (>95% purity) - Extended TFF membrane lifetime by 30% through process optimization	

PROJECTS

Defect Chemistry of RP Phase $(\text{La, Nd, Pr})_2\text{NiO}_{4+\delta}$ Study	2024 to 2026
- Performed DFT calculations on certain oxide RP phase perovskites. - Calculated the defect formation energy, density of states (DOS), phonon density of states (PhDOS), elastic tensor, dielectric tensor, defect free energy and chemical potential phase diagram.	
Bio-based Asphalt Molecular Dynamics Study	2020 to 2022
- Performed classical molecular dynamics simulations to characterize temperature-dependent transport behavior - Extracted viscoelastic properties across multiple temperature regimes to inform material performance	
Carbon Nanotube Purification Study	2019 to 2020
Designed and implemented purification protocols achieving 95% nanotube homogeneity	

EDUCATION

M.S. Materials Science & Engineering Worcester Polytechnic Institute, Worcester, MA	GPA: 3.50 2024 to 2026
B.S. Chemical Engineering & B.S. Applied Mathematics University of Rhode Island, Kingston, RI Minors: Chemistry, Physics	GPA: 3.16 2019 to 2022

TECHNICAL SKILLS

Atomistic Simulation:	Density Functional Theory (VASP & GPAW), Molecular Dynamics (LAMMPS)
Scientific Computing:	Python, MATLAB, ASPEN Plus Dynamics, Linux, LaTeX
Theory & Modeling:	Thermodynamics, Transport Phenomena, Defect Chemistry
Laboratory:	Glovebox operation, HPLC, AKTA FPLC, TFF, SEM/XRD, DSC/TGA
Process Engineering:	Manufacturing optimization, SOP development, statistical process controls